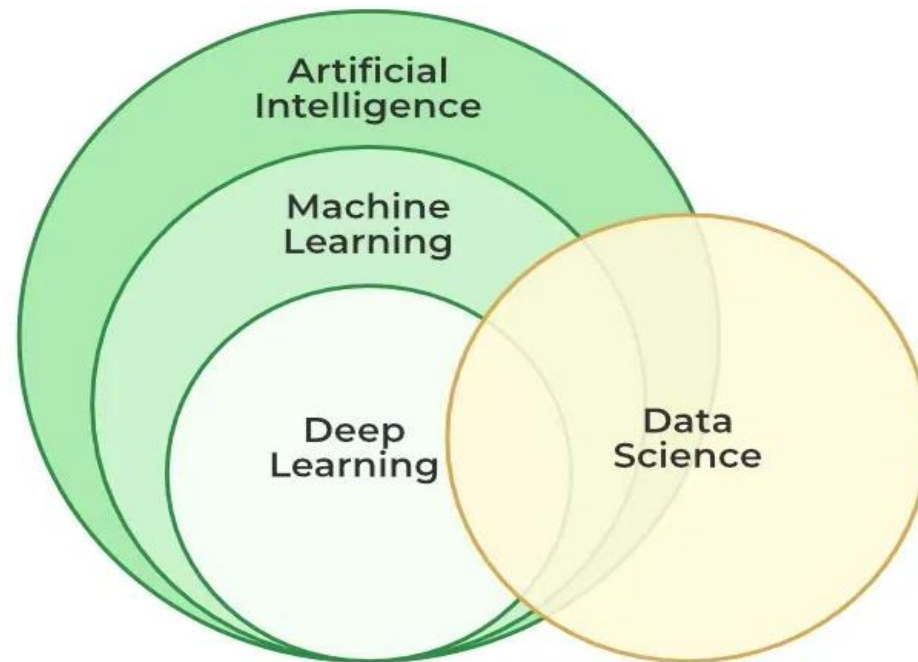


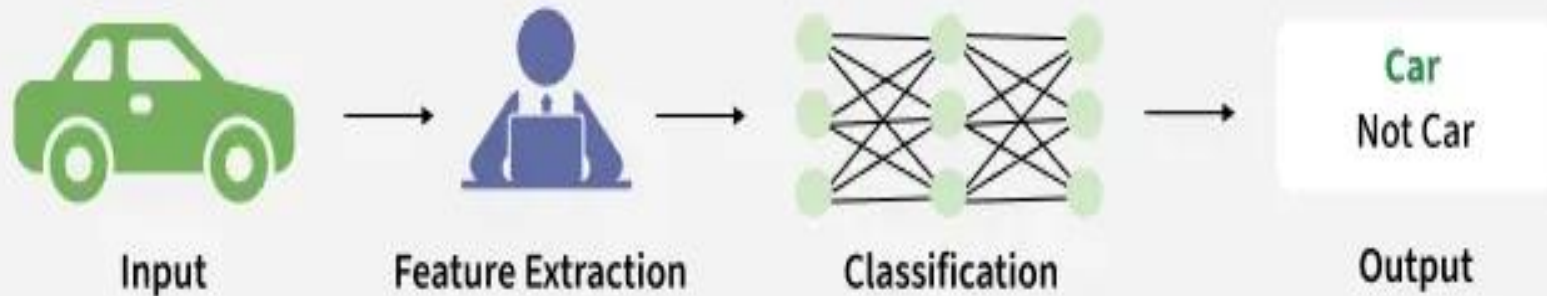
Unit I: Introduction to Deep Learning

Difference between Machine Learning and Deep Learning

Machine learning and Deep Learning both are subsets of artificial intelligence but there are many similarities and differences between them.

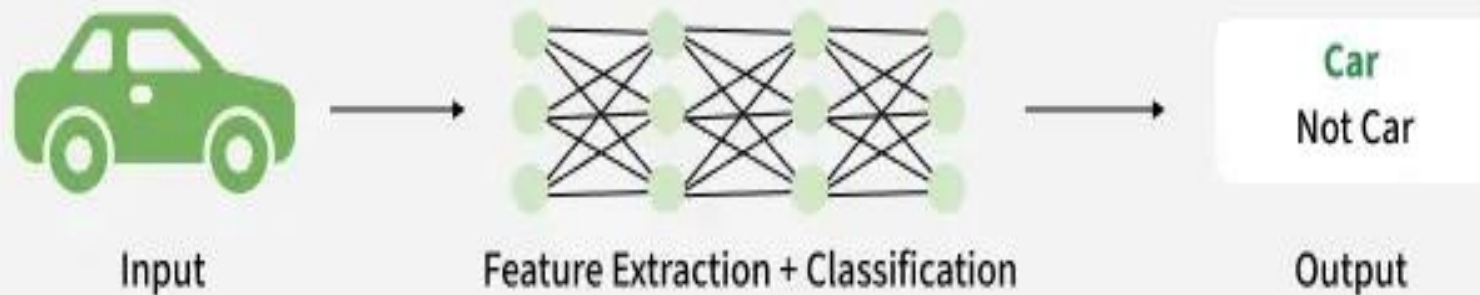


Machine Learning



Uses algorithms and learns on its own but may need human intervention to correct errors

Deep Learning



Uses advanced computing, its own neural network, to adapt with little to no human intervention

Machine Learning	Deep Learning
Apply statistical algorithms to learn the hidden patterns and relationships in the dataset.	Uses artificial neural network architecture to learn the hidden patterns and relationships in the dataset.
Can work on the smaller amount of dataset	Requires the larger volume of dataset compared to machine learning
Better for the low-label task.	Better for complex tasks like image processing, natural language processing, etc.
Takes less time to train the model.	It takes more time to train the model.
A model is created by relevant features which are manually extracted from images to detect an object in the image.	Relevant features are automatically extracted from images. It is an end-to-end learning process.

Less complex and easy to interpret the result.

It can work on the CPU or requires less computing power as compared to deep learning.

More complex, it works like the black box interpretations of the result are not easy.

It requires a high-performance computer with GPU.

History And Evolution Of Neural Networks

